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GCSE · OCR · Computer Science

1 hour 16 questions

CPU Architecture, Performance & Embedded Systems

CPU Architecture, Performance & Embedded Systems

The Purpose of the CPU / CPU Components & Their Function / Von Neumann Architecture / Characteristics of the CPU / Embedded Systems

Easy (5 questions)	/15
Medium (9 questions)	/37
Hard (2 questions)	/12
Total Marks	**/64**

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Easy Questions

1 (a) The specification of two CPUs is shown in **Fig. 1**.

Computer 1	Computer 2
Clock Speed: 1 GHz	Clock Speed: 1.4 GHz
Cache size: 2 MB	Cache size: 2 MB
Number of Cores: 4	Number of Cores: 2

Fig. 1

When running a 3D flight simulator, **Computer 1 is likely to run faster than Computer 2**.

Using the information in Fig. 1, **identify one reason** for this.

..... **(1 mark)**

(b) **Identify two** other parts of a computer that are not in **Fig. 1**, which could **improve the performance** of the computers.

.....

..... **(2 marks)**

2 **Identify two events** that take place during the **fetch-execute cycle**.

.....

..... **(2 marks)**

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3 (a) A Sat Nav contains an embedded system.

Define what is meant by an '**embedded system**'.

..... **(1 mark)**

(b) **Identify three devices**, other than a Sat Nav, which contain embedded systems.

.....

.....

..... **(3 marks)**

4 Von Neumann architecture includes **registers**.

Identify two registers used in Von Neumann architecture.

.....

..... **(2 marks)**

5 **Match** the letter in the definitions to the part of the processor it describes

Definitions

A	Performs mathematical operations
B	Sends signals to direct the operations
C	Keeps the clock in sync
D	A small piece of memory inside the processor that can hold one instruction or address
E	High speed memory inside the processor that stores recently used instructions

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Part of the processor

Control Unit (CU)	
Cache	
Arithmetic Logic Unit (ALU)	
Register	

(4 marks)

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Medium Questions

1 Complete the table by writing the **missing definition** or **name of** each of the common **CPU components** and **registers**.

CPU component or register	Definition
	Stores the address of the next instruction to be fetched from memory. Increments during each fetch-execute cycle.
CU (Control Unit)	
	Stores the address of the data to be fetched from or the address where the data is to be stored.
	Performs mathematical calculations and logical operations.

(4 marks)

2 Explain **one** reason why the **cache size** affects the **performance of the CPU**.

.....

..... **(2 marks)**

3 (a) A computer system has a 2.5 GHz processor and 5 GB of RAM.

State the **purpose** of a CPU.

..... **(1 mark)**

(b) **State** what is meant by a **single core 2.5 GHz** processor.

.....

..... **(2 marks)**

4 Ali's **tablet computer** has an operating system.

Ali thinks his tablet is an **embedded system**.

State whether Ali is **correct or incorrect**, **justifying your choice**.

.....

.....

..... **(3 marks)**

5 (a) Xander has a **smart watch** and a **laptop**.

Tick (✓) one box to show whether the smart watch or the laptop is an example of an **embedded system**.

	Is an example of an embedded system
Smart watch	
Laptop	

(1 mark)

(b) **Justify** your choice to part (**a**).

.....

..... **(2 marks)**

6 (a) Alicia has designed a computer using **Von Neumann architecture**.

Describe the purpose of **two registers** that are used by Von Neumann architecture.

.....

.....

.....

..... **(4 marks)**

(b) The CPU has a **clock speed** of 3.8GHz.

Describe what is meant by a **clock speed of 3.8GHz**.

.....

..... **(2 marks)**

7 (a) A computer has a Central Processing Unit (CPU).

Describe what happens during the fetch-execute cycle.

.....

..... **(2 marks)**

(b) Name **two** registers used in the fetch-execute cycle and the purpose of each register.

.....

.....

.....

..... **(4 marks)**

(c) Give **three** characteristics of a CPU that can affect its performance.

.....

.....

..... **(3 marks)**

8 A car has a 'Follow Me' system that uses a cruise control feature to allow the car to follow the car in front of it. It will keep the same speed and distance without the driver's intervention. The cruise control system is an example of an embedded system.

Explain the reasons why the 'Follow Me' system is an example of an embedded system.

.....

.....

..... **(3 marks)**

9 Complete the table by writing the **missing definition** or **name of** each of the common **CPU components** and **registers**.

CPU component or register	Definition
	Temporarily stores data and instructions that are frequently accessed by the CPU to reduce access time to main memory.
ALU (Arithmetic Logic Unit)	
	A small, high-speed storage location within the CPU that holds data temporarily during processing.
	Manages and coordinates all CPU operations by sending control signals to other components.

(4 marks)

Hard Questions

1 (a) Kerry wants to buy a new computer, but she does not understand what the different parts of a computer do.

Kerry has heard of a **CPU** but does not know what it is.

The following sentences describe the purpose of a CPU.

Complete the sentences by **filling in the missing words**.

> **CPU stands for** .....
>
> **It is the part of the computer that fetches and executes the**
>
> ..... **that are stored in** .....**.**
>
> **The CPU contains the Arithmetic** ..... **Unit (ALU) and the**
>
> ..... **Unit (CU).**

(5 marks)

(b) Kerry is looking at two computers; one has a single core processor and the other has a dual core processor.

Explain why having a **dual core** processor might **improve the performance** of the computer.

.....

..... **(2 marks)**

(c) One computer has 64 kilobytes of cache and the other has 512 kilobytes of cache.

Explain how the **cache size** can **affect the performance** of the CPU.

.....

..... **(2 marks)**

2 Alicia says:

"*My computer has a quad-core processor, so it will run twice as fast as a computer with a dual-core processor.*"

Explain why this statement **is not always true**.

.....

.....

..... **(3 marks)**